

RECURSION

* PATTERN QUESTION

$n = 4$

```

* * * *
* * *
* *
*
  
```

$f(\text{row}, \text{column})$

$\text{row} = 4 \quad \text{column} = 0$

```

4, 0 print *
↓
4, 1 p "
↓
4, 2 "
↓
4, 3 "
↓
  newline *
↓
3, 0 "
↓
: "
↓
: "
  
```

when :-

$\text{row} == \text{column}$

$\text{row}--$ & next line

* Pattern 2 $n = 4$

```

* *
* * *
* * * *
  
```

$f(\text{row}, \text{column})$

$\text{row} = 3 \quad \text{column} = 0$

```

4, 0 print*
↓
4, 1 print*
↓
4, 2 print*
↓
4, 3 print*
↓
3, 0 next line print*
↓
: print*
↓
:
  
```

Here: first all function will be called then it will start printing from back

Bubble Sort Using Recursion :-

• arr [4, 3, 2, 1] row = 4
column = 0 [4, 3, 2, 1]
if $c > c+1$ c c+1
swap 3, 4, 2, 1
c c+1
c++; 0, 1
~~3, 2, 4~~ 3, 2, 4, 1
column < row c c+1
3, 2, 1, 4

When row == column

then row →

column = 0

Base Case :-

row == 0 , return;

* Selection Sort. $\Rightarrow$ select max & put at last position

arr = [4, 3, 2, 1]

row = 4, column = 0

max = 0; max = 0

max = 0

arr[c] > arr[max] $4 > 4 \times$

when column < row

max = 0

c > ~~arr~~ max

[4, 3, 2, 1]

max = c

c = 1

call function with max

arr[c] > arr[max] $3 > 4 \times$

as parameter included

max = 0

& c++

[4, 3, 2, 1]

c = 2

$2 > 4 \times$

when column = row

max = 0

Swap max with row.

[4, 3, 2, 1]

c = 3 $1 > 4 \times$

then

Now

call function again

c = row

row = 4 max = 0

& after Base case

swap

row == 0

[1, 3, 2, 4]

return

!

function $\Rightarrow$ function (arr, row, col, max).